



UNIVERSITÉ DE
RENNES 1



CVS
CACHAN
BRETAGNE

Reduction of the worst-case execution time of real-time tasks targeting many-core platforms

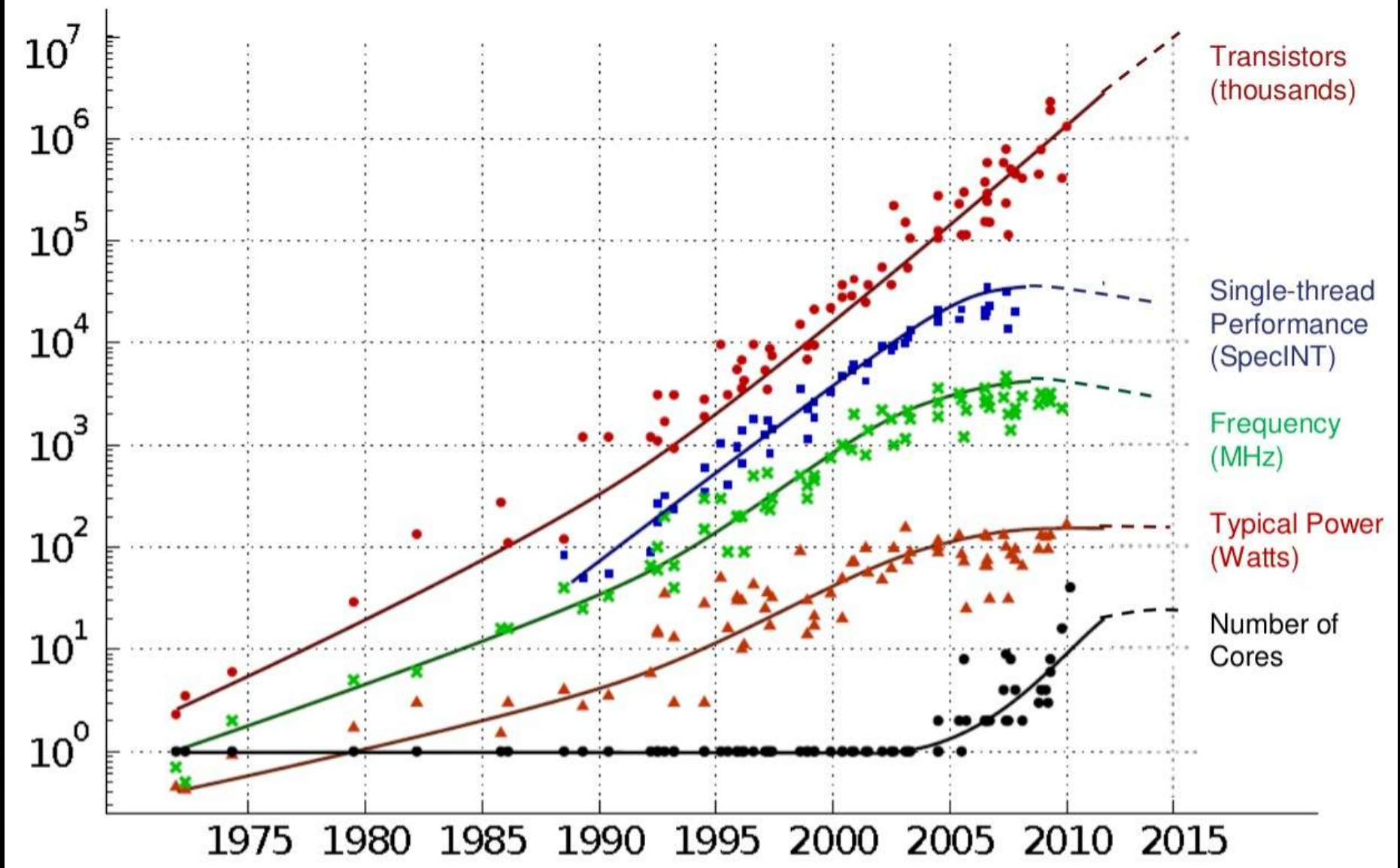
Student: Bastien Pasdeloup
Director: Eduardo Tovar
Supervisor: Vincent Nélis

2015

... saving power

Designing
transformations
conclusion





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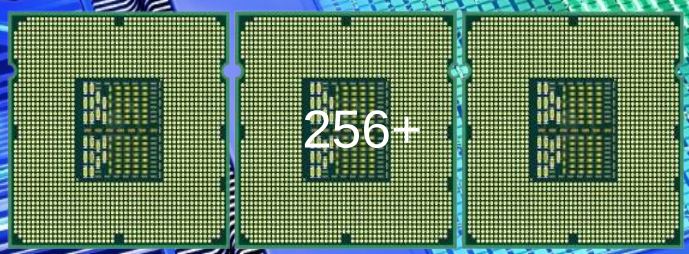
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ns

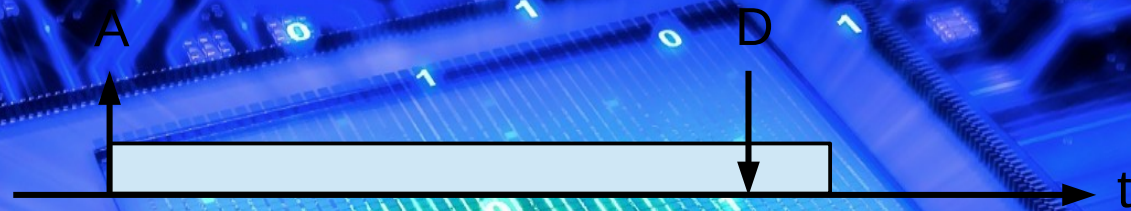
Conclu

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Programs

Dependencies
Transformations
Analysis
Conclusions



WCE

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transformations

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conclusion

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KALRAY

MPPA
— MANYCORE

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TWNTFYWW

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t

- $\text{var}_d := \text{var}_{s1} [\bullet \text{var}_{s2}]$
- Every instruction executes in one unit of time
- No loops or recursion
- Infinite number of cores
- No overheads

the call

transformations

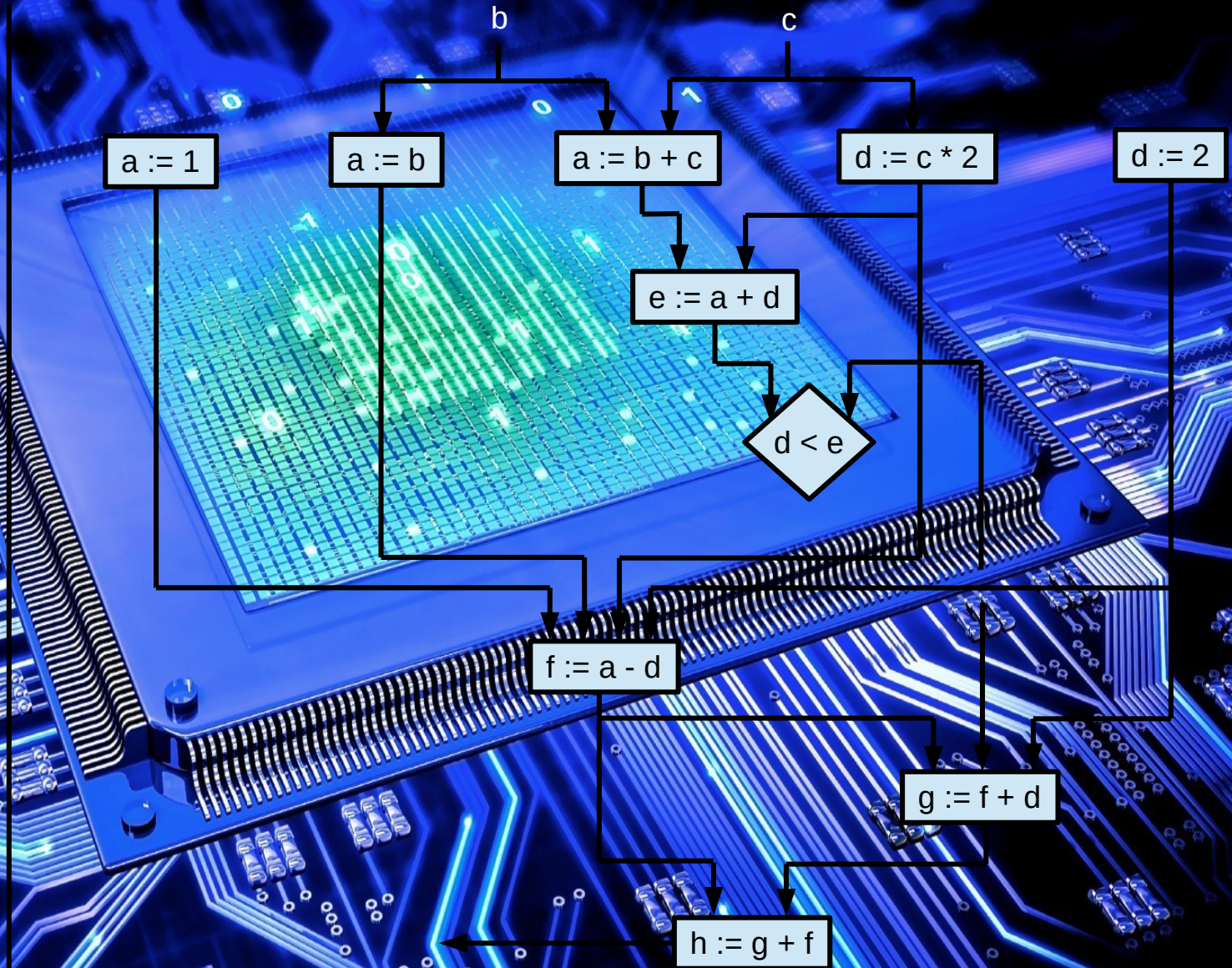
conclusion

- Data flow graph
 - Data dependencies graph
 - The ambiguous entries
 - Insertion of the selectors
- Transformations
 - Time loss analysis
 - Reduction of the time loss

function f(b, c)

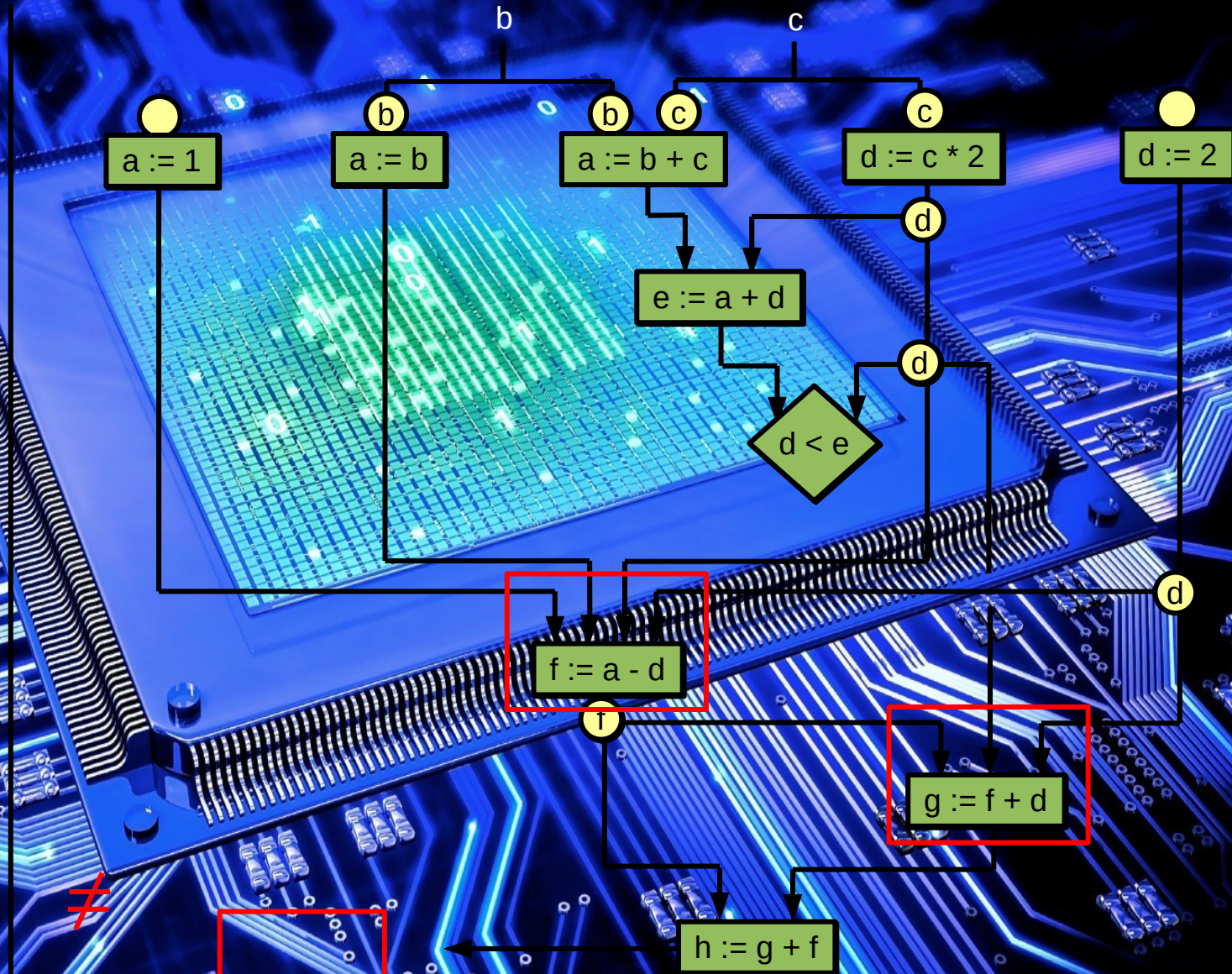
```

a := b + c
d := c * 2
e := a + d
if (d < e)
  a := 1
else
  a := b
  d := 2
f := a - d
g := f + d
h := g + f
  
```



function (c)

```
a := b + c  
d := c * 2  
e := a + d  
if (d < e)  
  a := 1  
else  
  a := b  
  d := 2  
  f := a - d  
  g := f + d  
  h := g + f
```



Problem

Goal

Transformations

Proof

Conclusion

$a := 1$

$a := b$

$d < e$

$d := c * 2$

$d := 2$

$a ?$

$d ?$

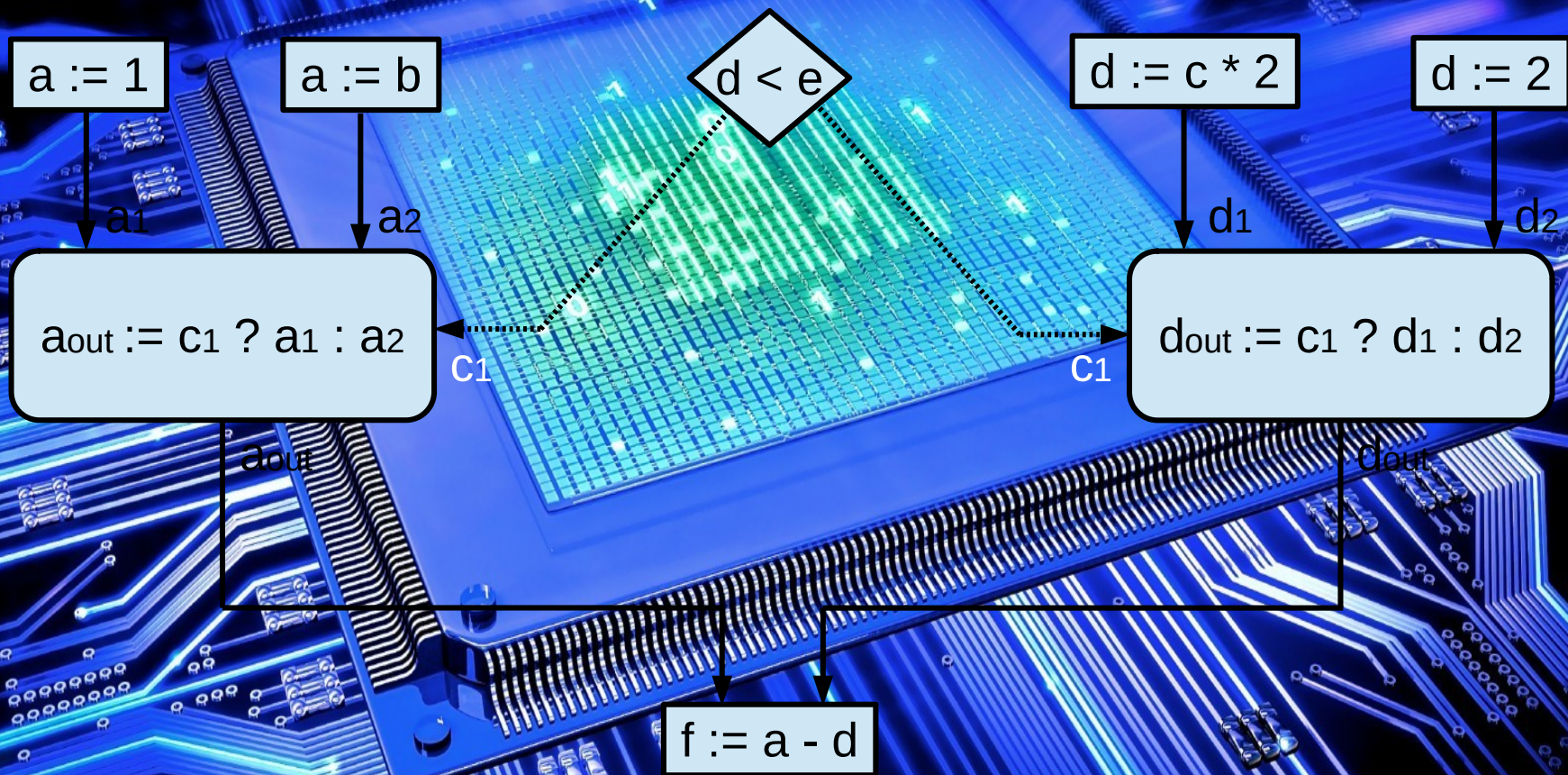
$f := a - d$

Problem

Data

Transformations

Conclusion

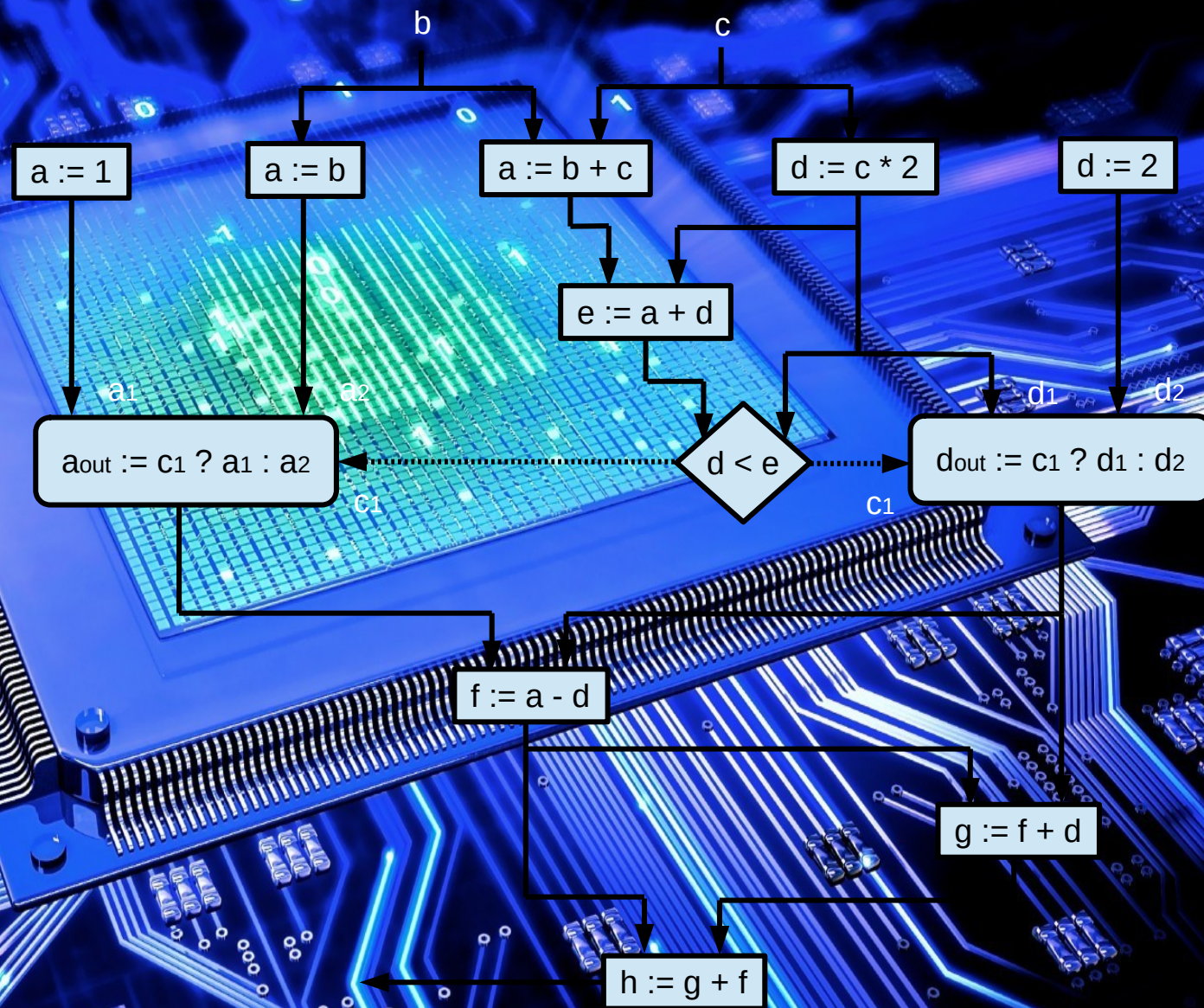


Control flow graph

Data flow graph
Transformations
Conclusion

function f(c)

```
a := b + c  
d := c * 2  
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if (d < e)  
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else  
  a := b  
  d := 2  
  f := a - d  
  g := f + d  
h := g + f
```



user-allocated

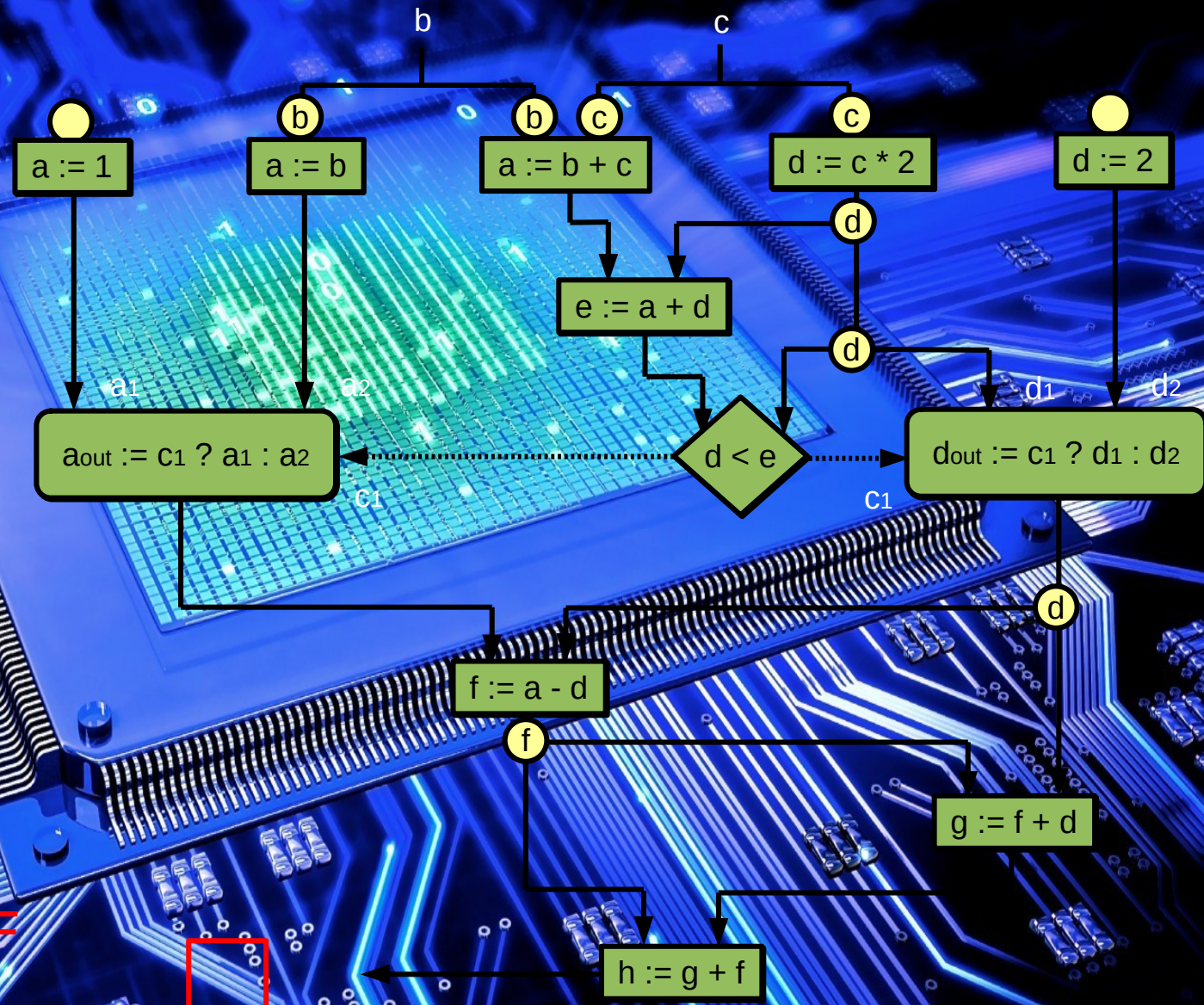
data

transformations

conclusion

function (c)

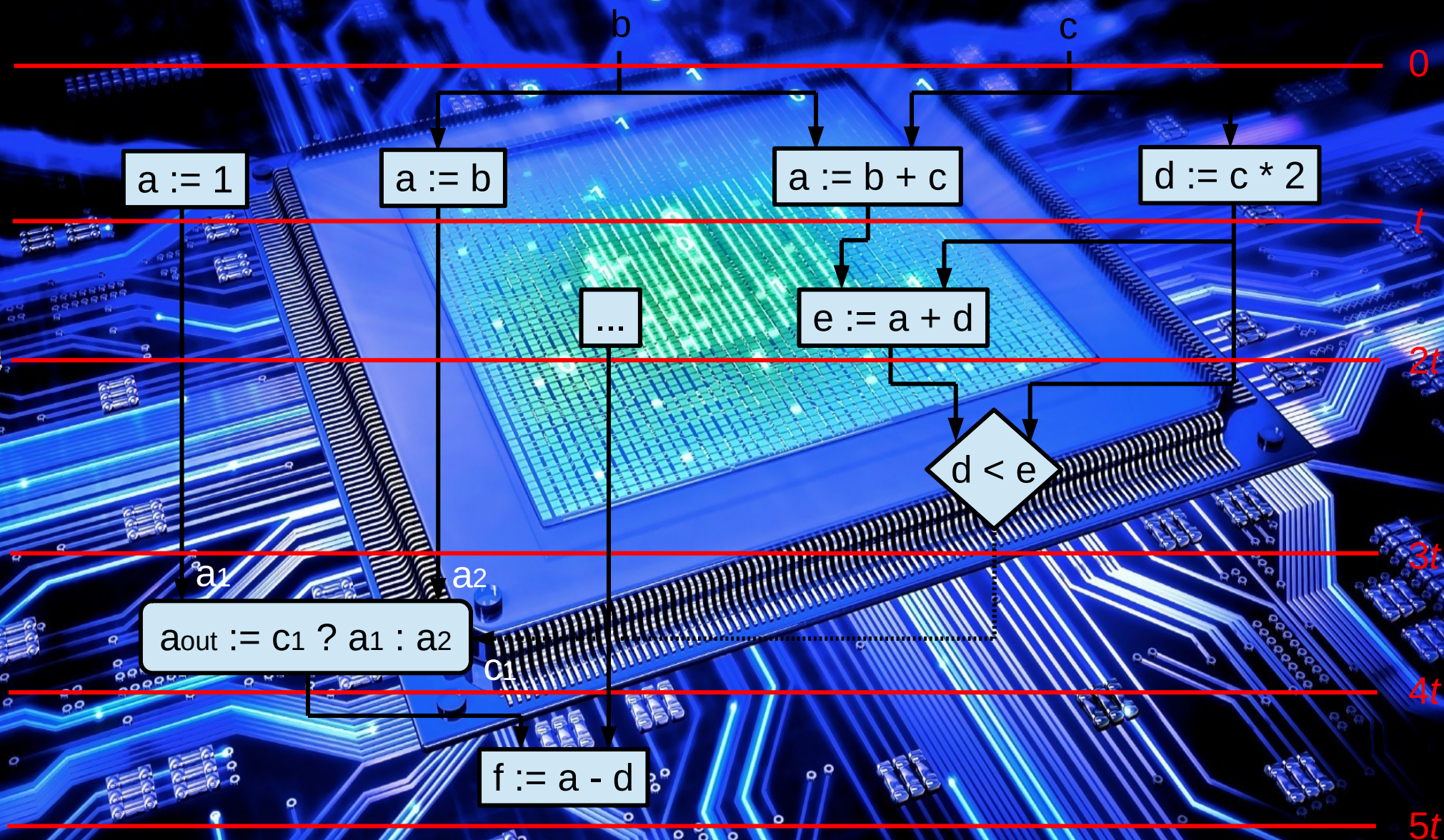
```
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e := a + d
if (d < e)
  a := 1
else
  a := b
  d := 2
  f := a - d
  g := f + d
  h := g + f
```



Input information

Decision points

Conclusion

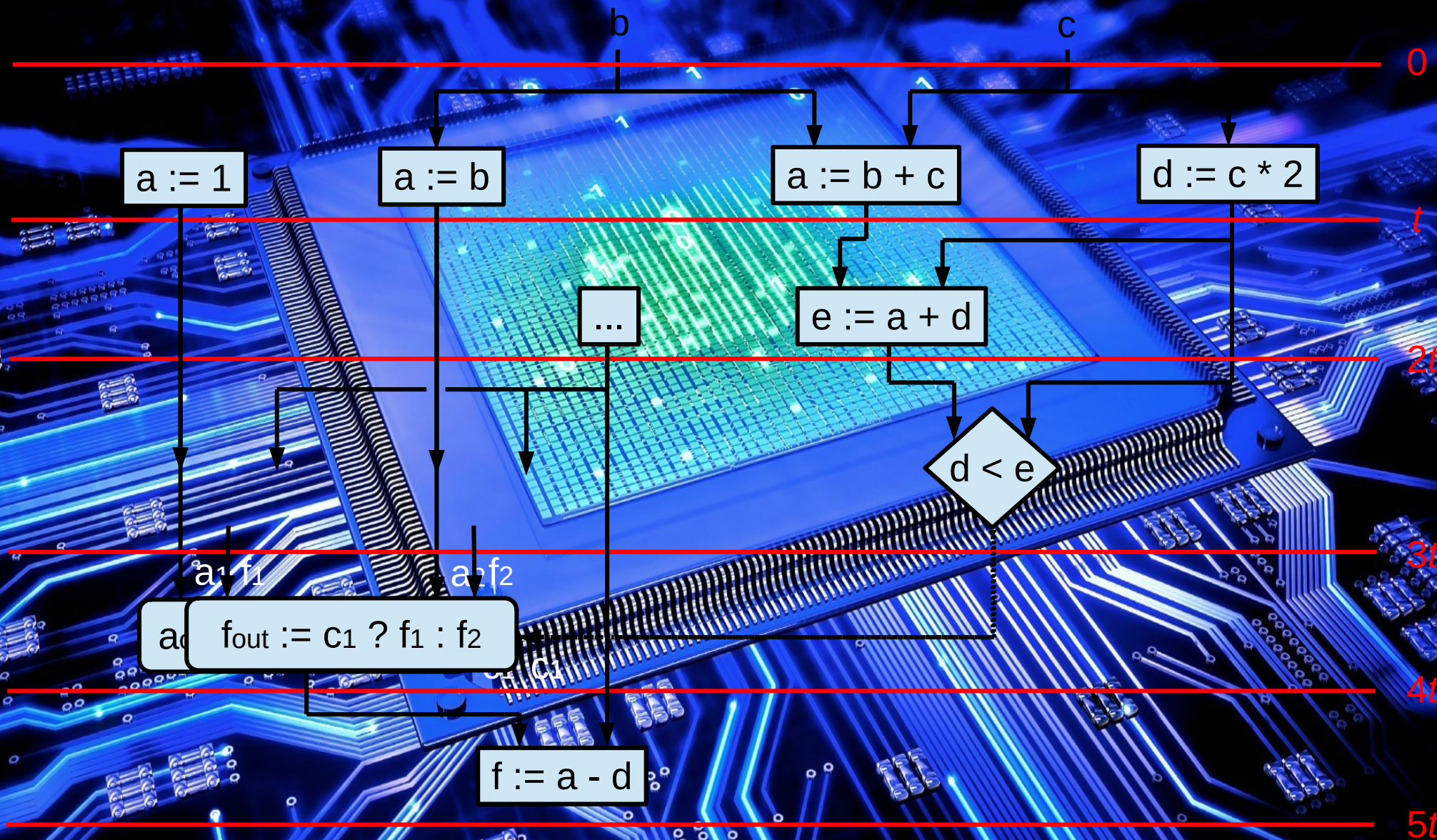


the code

D

transformations

conclusion

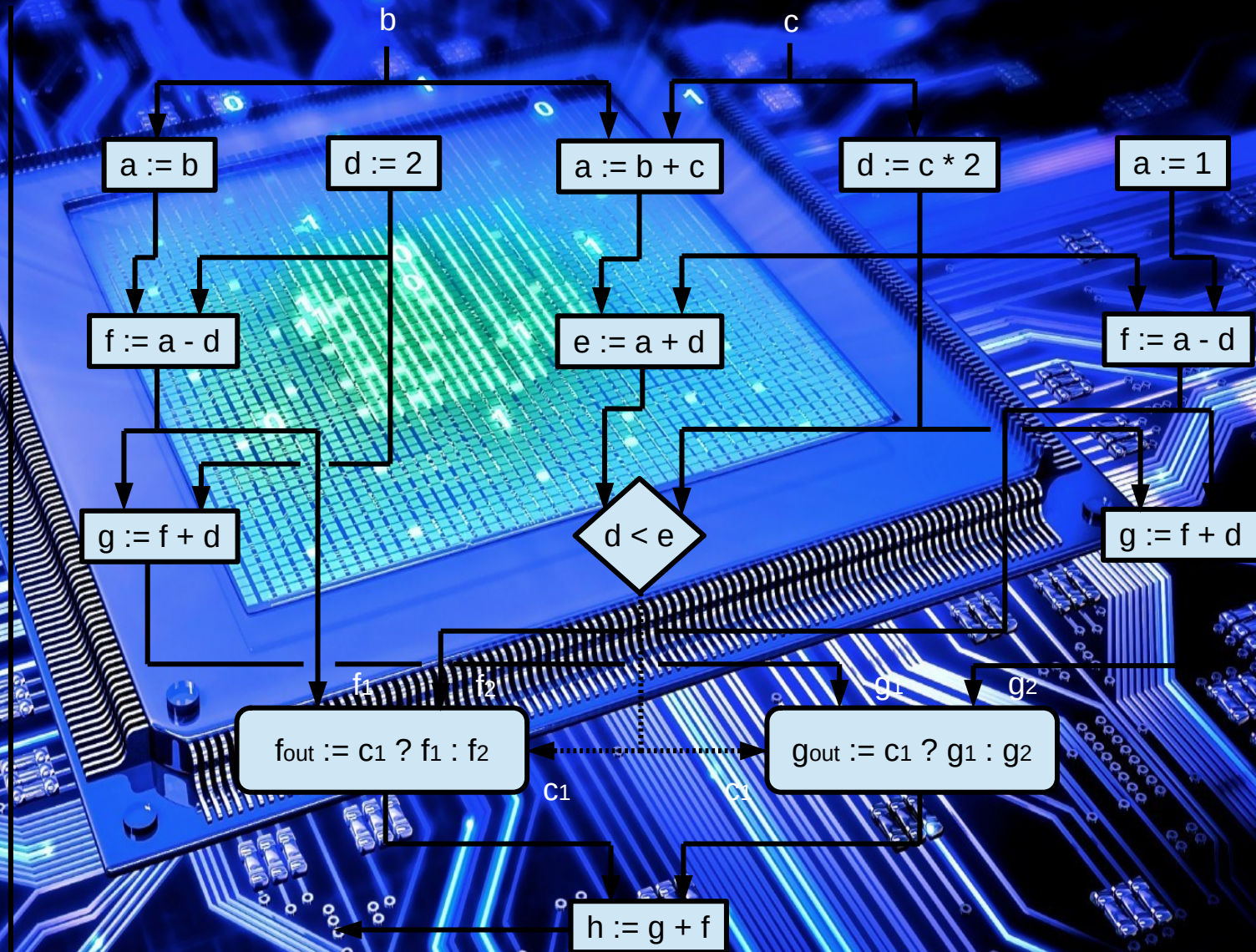


Flow graph

Decision points
Conclusion

Function (c)

```
a := b + c  
d := c * 2  
e := a + d  
if (d < e)  
  a := 1  
else  
  a := b  
  d := 2  
  f := a - d  
  g := f + d  
  h := g + f  
end if
```

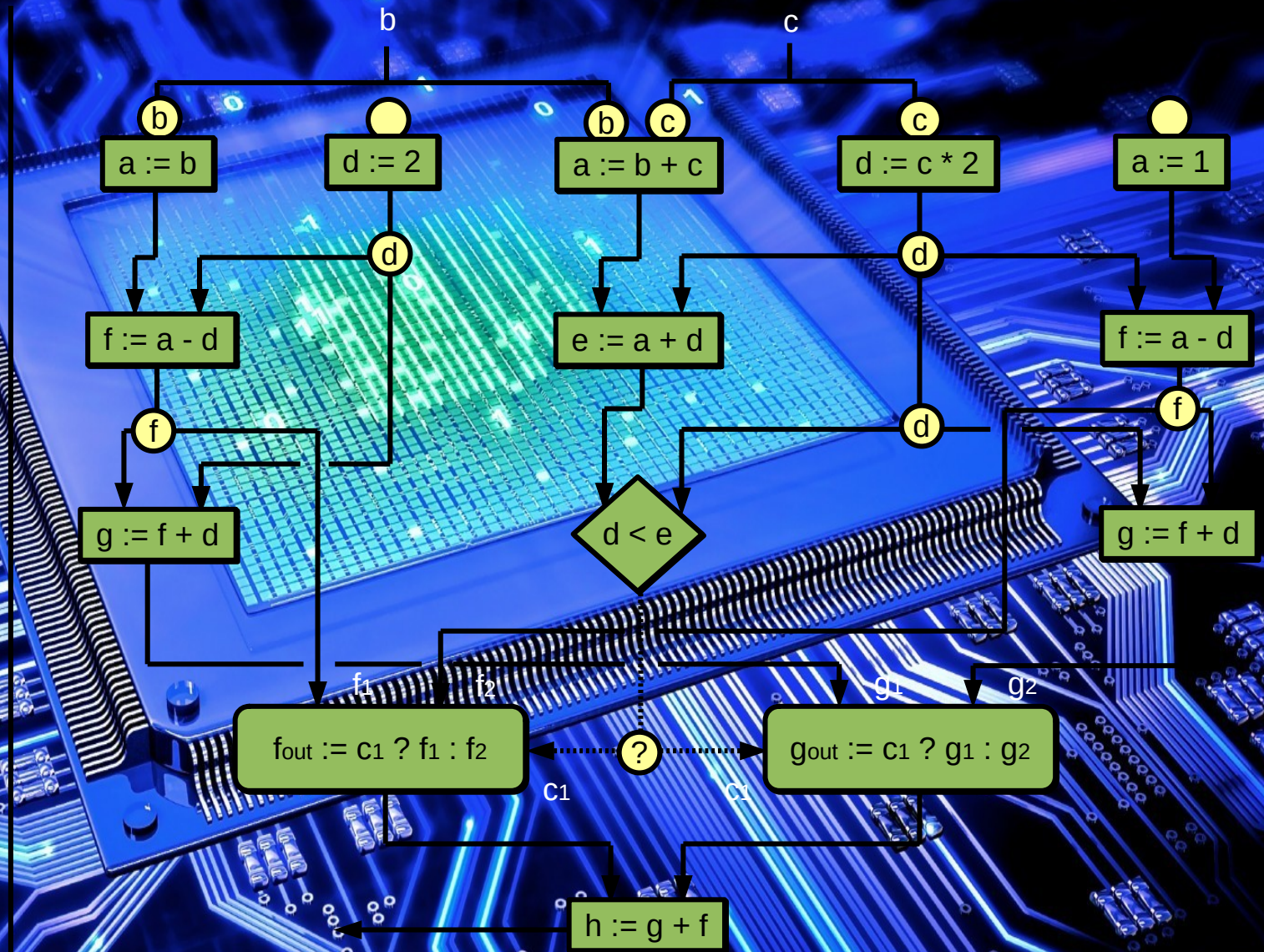


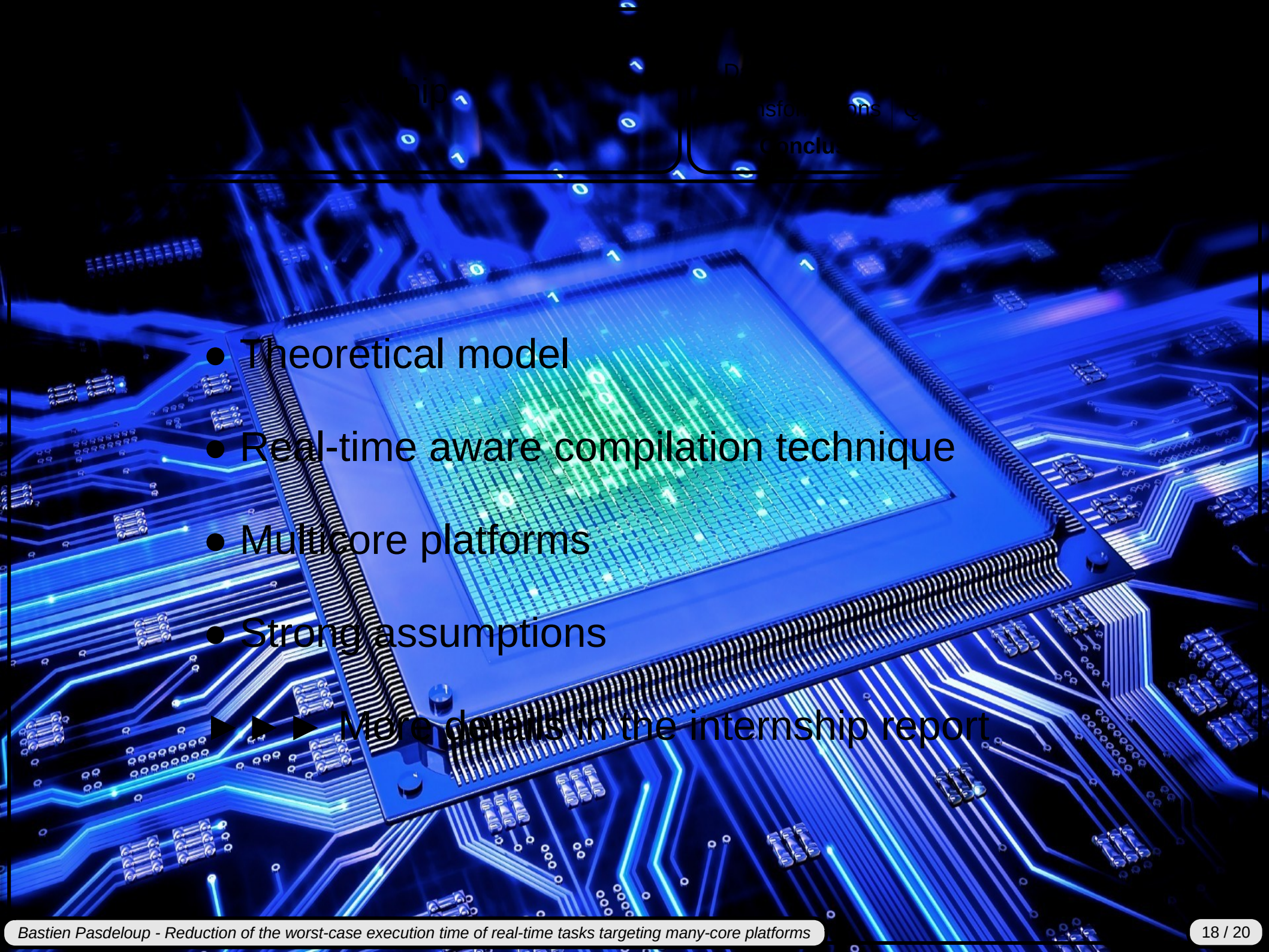
program

Disjunctions
Conjunctions

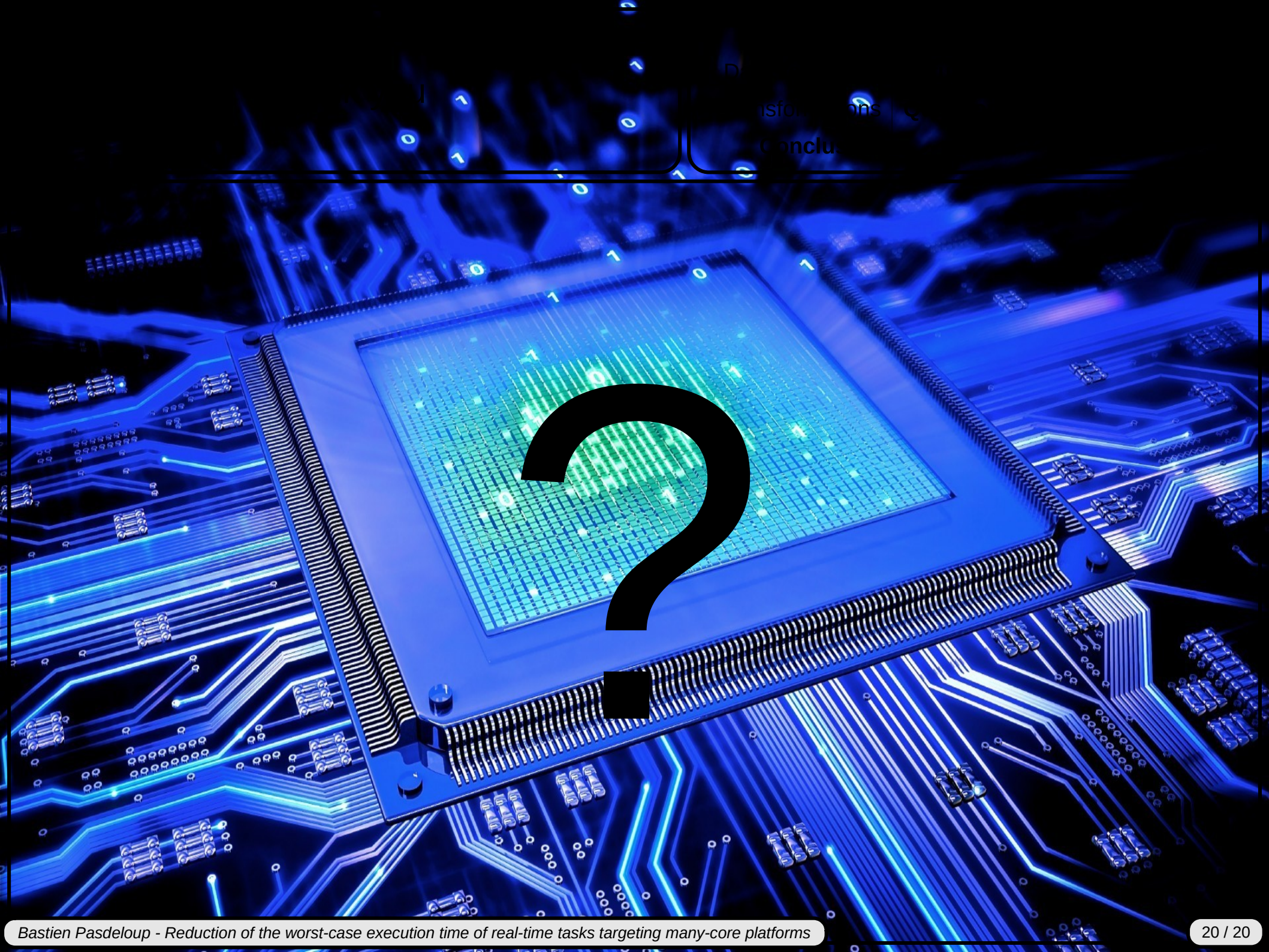
function (c)

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a := b + c  
d := c * 2  
e := a + d  
if (d < e)  
  a := 1  
else  
  a := b  
  d := 2  
  f := a - d  
  g := f + d  
  h := g + f
```



- 
- Theoretical model
 - Real-time aware compilation technique
 - Multicore platforms
 - Strong assumptions
- ▶▶▶ More details in the internship report

- $\text{var}_d := \text{var}_{s1} [\bullet \text{var}_{s2}]$
- Every instruction executes in one unit of time
- No loops or recursion
- Infinite number of cores
- No overheads



Dispositions | Q
conclus